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5. Simplifying the top and bottom, our single fraction is six over thirty-five.

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Example. $\frac{x-3}{x-6} \cdot \frac{x-1}{x+6}$.

1. Let's look at another example. What is x minus three over x minus six times x minus one over x plus six? Note that the 3 and 6 do not cancel in the first fraction. In fact none of the x's cancel in any fraction either.
2. As a reminder, for multiplying fractions, there is no need for a common denominator.
3. To multiply these fractions, we multiply the numerators together and the denominators together. The previous example was fractions with numbers only. We *still* apply this process for fractions with variables.
4. Simplifying the top and bottom, we get x squared minus 4 x plus 3 over x squared plus 6 x minus 6 x minus 36.
5. On bottom, the 6 x and negative 6x undo each other. We're left with x squared minus 4 x plus 3 over x squared minus 36.

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Example. $\frac{x-5}{x^2-100} + \frac{2}{x-10}$.

1. Let's look at another example. What is x minus 5 over x squared minus 100 plus two over x minus 10?
2. To simplify adding fractions, we need a common denominator, just like we would if we had fractions without variables. Our work becomes easier if we first notice that the first denominator factors as x plus 10 times x minus 10.
3. Let's rewrite it like this.
4. Then, we can leave the first fraction alone. For the second fraction, multiply the top and bottom by x plus 10.
5. As we do this, the second fraction needs the original x minus 10 in parentheses. Without parentheses, the Order of Operations says something else.
6. Now that we have a common denominator, we can combine the two fractions into one. The numerator is x minus 5 plus 2 times the quantity x plus 10. The denominator is x plus 10 times x minus 10.
7. Distribute the 2 in the numerator. The denominator is x squared minus 100.
8. We end with $3x + 15$ over x squared minus 100. No further cancellations are possible because the last operations on top and bottom are plus and minus, not times.

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Example. $\frac{x-5}{x^2-100} \div \frac{2}{x-10}.$

1. Let's look at another example. What is x minus 5 over x squared minus 100 divided by two over x minus 10?
2. Leave the first fraction alone, and reciprocate the second fraction at the *same* time that we change the division sign into multiplication.
3. Since we are now multiplying fractions, multiply straight across, with no need for a common denominator. We have the quantity x minus 5 times the quantity x minus 10 on top, and the quantity x squared minus 100, close parentheses, times 2 on bottom. *All* parentheses here are needed due to order of operations.
4. Expanding everywhere is an option, but that takes us further from a factored form. Instead, we factor the x squared minus 100 on bottom as x plus 10 times x minus 10. Now, everything is a factor, so any common factor on top and bottom will cancel.
5. Cancelling the common factor of x minus 10 on top and bottom, we are left with x minus 5 over 2 times the quantity x plus 10. Since nothing further will cancel, it would be fine to leave the denominator like this or to expand by distributing 2.

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